

Capture the Flag

Your Student Portal is now dynamic thanks to interpolation.

But there's a problem.

Your announcement component currently has something like:

```
announcementTitle1 = "Enrollment is Now Open";  
announcementBody1 = "Enrollment for the next semester is now  
available.";
```

```
announcementTitle2 = "Intramurals 2026";  
announcementBody2 = "Registration is now open.";
```

```
announcementTitle3 = "Library Schedule";  
announcementBody3 = "The library will be open until 8 PM.";
```

And your HTML has to repeat the same card structure three times.

The school now has **20 announcements**.

Your mission:

Stop repeating yourself. Make Angular generate the cards for you.

Flag 0 – Mission Branch

Create and switch to a new branch: `feature/ngFor`



Capture

- Show your current Git branch.

```
Student@COLLEGEPC-9 MINGW64 ~/my-first-angular-app (feature/ngFor)
$ git switch -c feature/ngFor
fatal: a branch named 'feature/ngFor' already exists

Student@COLLEGEPC-9 MINGW64 ~/my-first-angular-app (feature/ngFor)
$ git switch feature/ngFor
D      src/Activity/Activity 1 - Card Replication.pdf
D      src/Activity/Activity 3_ Interpolation (1).pdf
D      src/Activity/Card Replication.pdf
D      src/Activity/ITFE403A_Activity #1.pdf
D      src/Activity/ITFE403A_Activity #2.pdf
Already on 'feature/ngFor'

Student@COLLEGEPC-9 MINGW64 ~/my-first-angular-app (feature/ngFor)
$
```

FLAG 1 — The Repetition Problem

Open your announcement component.

Look at your existing announcement cards.

You should currently have something similar to:

```
<div class="announcement-card">
  <h3>{{ announcementTitle1 }}</h3>
  <p>{{ announcementBody1 }}</p>
</div>
```

```
<div class="announcement-card">
  <h3>{{ announcementTitle2 }}</h3>
  <p>{{ announcementBody2 }}</p>
</div>
```

```
<div class="announcement-card">
  <h3>{{ announcementTitle3 }}</h3>
  <p>{{ announcementBody3 }}</p>
</div>
```

Mission

Imagine the school gives you **10 more announcements**.

Would you want to copy and paste the card **10 more times**?

Don't change anything yet.



Capture: Your repeated announcement cards.

src > app > announcement-card > <> announcement-card.html >  div.announcement-card >  h3

```
1  <div class="announcement-card">
2    <h3>{{ announcementTitle1 }}</h3>
3    <p>{{ description1 }}</p>
4  </div>
5
6  <div class="announcement-card">
7    <h3>{{ announcementTitle2 }}</h3>
8    <p>{{ description2 }}</p>
9  </div>
10
11 <div class="announcement-card">
12   <h3>{{ announcementTitle3 }}</h3>
13   <p>{{ description3 }}</p>
14 </div>
15
16 <div class="announcement-card">
17   <h3>{{ announcementTitle4 }}</h3>
18   <p>{{ description4 }}</p>
19 </div>
20
21 <div class="announcement-card">
22   <h3>{{ announcementTitle5 }}</h3>
23   <p>{{ description5 }}</p>
24 </div>
25
26 <div class="announcement-card">
27   <h3>{{ announcementTitle6 }}</h3>
```

FLAG 2 — Put the Data Together

Instead of creating separate variables, create **one array**.

Research Hint

An array can contain multiple objects.

You don't need to memorize the syntax.

Search:

Angular array of objects



Capture: Your **announcements** array.

```
TS announcement-cards M X announcement-card.html M
src > app > announcement-card > TS announcement-cards > AnnouncementCard > announcements
1 import { Component, Input } from '@angular/core';
2
3 @Component({
4   selector: 'app-announcement-card',
5   standalone: true,
6   imports: [],
7   templateUrl: './announcement-card.html',
8   styleUrls: ['./announcement-card.css']
9 })
10 export class AnnouncementCard {
11
12   announcements: { title: string; description: string }[] = [
13     {
14       title: '📅 Enrollment Extended',
15       description: 'Enrollment for the 2nd Semester has been extended until August 15, 2026.'
16     },
17     {
18       title: '📖 Library Hours Updated',
19       description: 'The college library is now open from 7:00 AM to 8:00 PM on weekdays.'
20     },
21     {
22       title: '🌙 Bosconian Night 2026',
23       description: 'Join us for Bosconian Night on October 1, 2026. The theme is Old Money. Wear your finest classic and elegant attire as we celebrate an unforgettable evening.'
24     },
25     {
26       title: '🎓 Graduation Requirements Updated',
27       description: 'Please note that the graduation requirements for the 2026 batch have been updated. Please refer to the updated guidelines for more information.'
28     },
29     {
30       title: '💼 Job Opportunities Available',
31       description: 'Several job opportunities are available for students interested in internships and part-time positions. Please check the career services portal for more details.'
32     },
33     {
34       title: '🎉 Cultural Festivals Coming Soon',
35       description: 'Get ready for our annual cultural festivals! Mark your calendars for the exciting events celebrating diversity and unity on campus.'
36     },
37     {
38       title: '📅 Academic Calendar Updated',
39       description: 'The academic calendar for the 2026-2027 school year has been released. Please review the dates for important deadlines and events.'
40     }
41   ];
42 }
```

```
TS announcement-cards M X announcement-card.html M
src > app > announcement-card > TS announcement-cards > AnnouncementCard > announcements
10 export class AnnouncementCard {
12   announcements: { title: string; description: string }[] = [
27     {
28       description: 'Please note that the graduation requirements for the 2026 batch have been updated. Please refer to the updated guidelines for more information.'
29     },
30     {
31       title: '💼 Job Opportunities Available',
32       description: 'Several job opportunities are available for students interested in internships and part-time positions. Please check the career services portal for more details.'
33     },
34     {
35       title: '🎉 Cultural Festivals Coming Soon',
36       description: 'Get ready for our annual cultural festivals! Mark your calendars for the exciting events celebrating diversity and unity on campus.'
37     },
38     {
39       title: '📅 Academic Calendar Updated',
40       description: 'The academic calendar for the 2026-2027 school year has been released. Please review the dates for important deadlines and events.'
41     },
42     {
43       title: '📖 New Textbooks Available',
44       description: 'New textbooks for the upcoming semester are now available in the campus bookstore. Check the bookstore website for a complete list of required materials.'
45     },
46     {
47       title: '💼 Internship Opportunities',
48       description: 'Several internship opportunities are available for students in various fields. Please visit the career services office for more information and application details.'
49     },
50     {
51       title: '🎉 Homecoming Celebration',
52       description: 'Join us for our annual Homecoming Celebration on November 15, 2026. Show your school spirit and enjoy a day filled with fun activities and memorable moments.'
53     }
54   ];
55 }
```

FLAG 3 — Meet ***ngFor**

Now comes the important part.

You have:

```
announcements = [  
  { ... },  
  { ... },  
  { ... }  
];
```

How can Angular create one card **for every item** in that array?

Hint

Research:

**Angular ngFor*

You should discover syntax similar to:

```
*ngFor="let announcement of announcements"
```

Your mission is to use it on your announcement card.

Instead of having:

```
<div class="announcement-card">
```

repeated three times, you should have **one card template** that Angular repeats.



Capture:

- Your updated code.

```
src > app > announcement-card > TS announcement-card.ts > Announc
1  import { Component } from '@angular/core';
2  import { NgFor } from '@angular/common';
3
4  @Component({
5    selector: 'app-announcement-card',
6    standalone: true,
7    imports: [NgFor],
8    templateUrl: './announcement-card.html',
9    styleUrls: ['./announcement-card.css']
10  })
```

```
src > app > announcement-card > <> announcement-card.html > div.announcement-card
1  <div *ngFor="let announcement of announcements" class="announcement-card">
2    <h3>{{ announcement.title }}</h3>
3    <p>{{ announcement.description }}</p>
4  </div>
```

FLAG 4 — Display the Current Announcement

You now have a loop variable.

Inside the repeated card, display title and body:

Your browser should now display all eight announcements.

Important

You should have:

1 array

1 card template

8 displayed cards

Not:

8 arrays

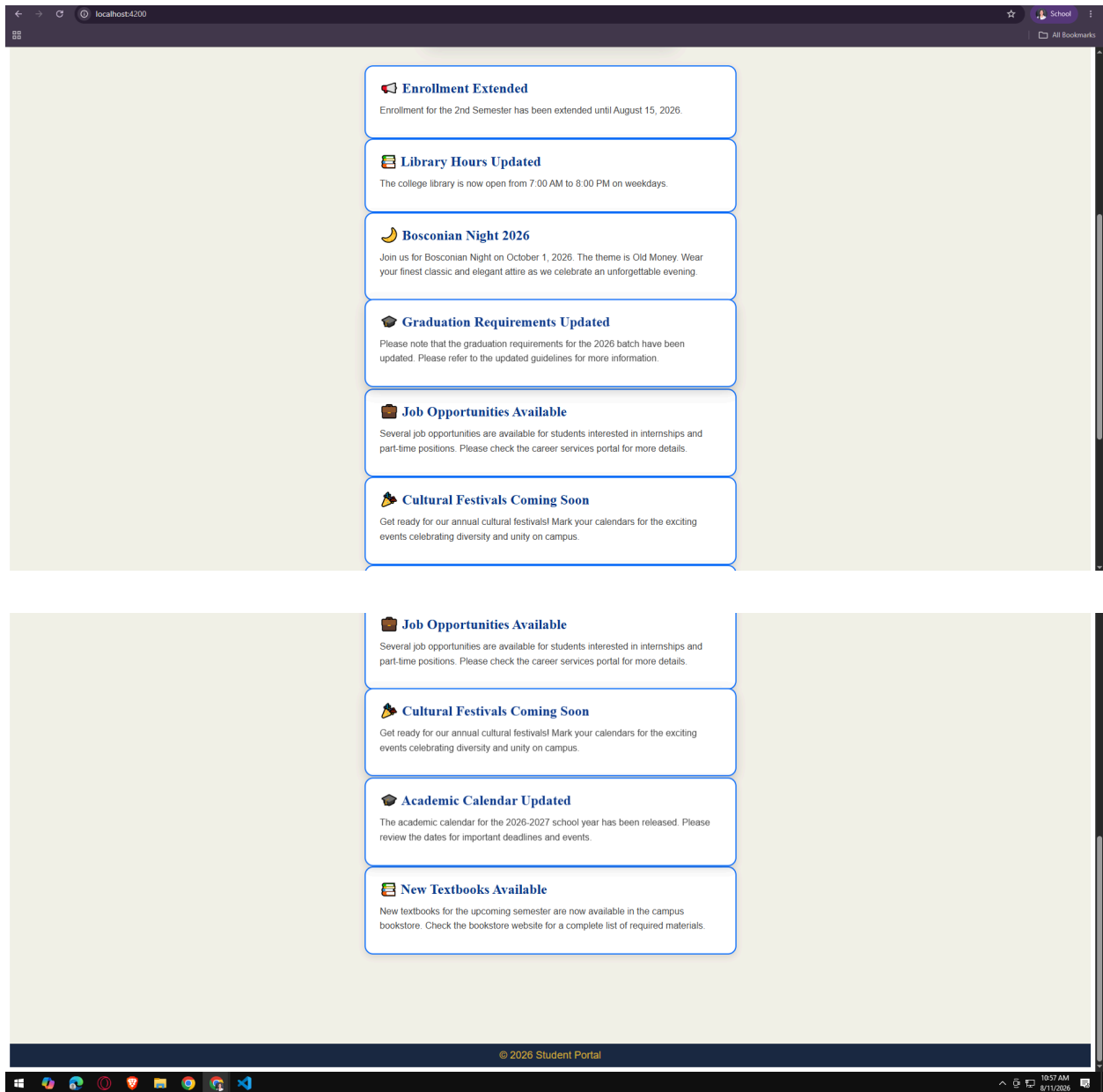
8 card templates



Capture:

- Browser showing all announcements.
- Updated code.

```
src > app > announcement-card > TS announcement-cards > AnnouncementCard
11 export class AnnouncementCard {
12
13   announcements: { title: string; description: string }[] = [
14     {
15       title: '📅 Enrollment Extended',
16       description: 'Enrollment for the 2nd Semester has been extended until August 15, 2026.'
17     },
18     {
19       title: '📖 Library Hours Updated',
20       description: 'The college library is now open from 7:00 AM to 8:00 PM on weekdays.'
21     },
22     {
23       title: '🌙 Bosconian Night 2026',
24       description: 'Join us for Bosconian Night on October 1, 2026. The theme is Old Money. Wear your finest classic and elegant attire as we celebrate an unforgettable evening.'
25     },
26     {
27       title: '🎓 Graduation Requirements Updated',
28       description: 'Please note that the graduation requirements for the 2026 batch have been updated. Please refer to the updated guidelines for more information.'
29     },
30     {
31       title: '💼 Job Opportunities Available',
32       description: 'Several job opportunities are available for students interested in internships and part-time positions. Please check the career services portal for more detail.'
33     },
34     {
35       title: '🎉 Cultural Festivals Coming Soon',
36       description: 'Get ready for our annual cultural festivals! Mark your calendars for the exciting events celebrating diversity and unity on campus.'
37     },
38     {
39       title: '📅 Academic Calendar Updated',
40       description: 'The academic calendar for the 2026-2027 school year has been released. Please review the dates for important deadlines and events.'
41     }
42   ]
43 }
```

FLAG 5 — Add More Data

Now test whether your loop actually works.

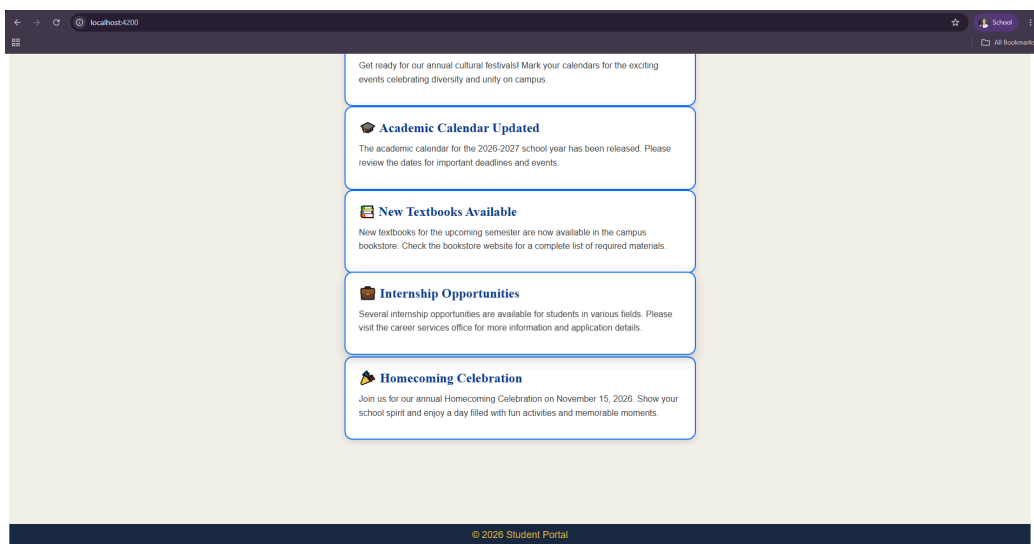
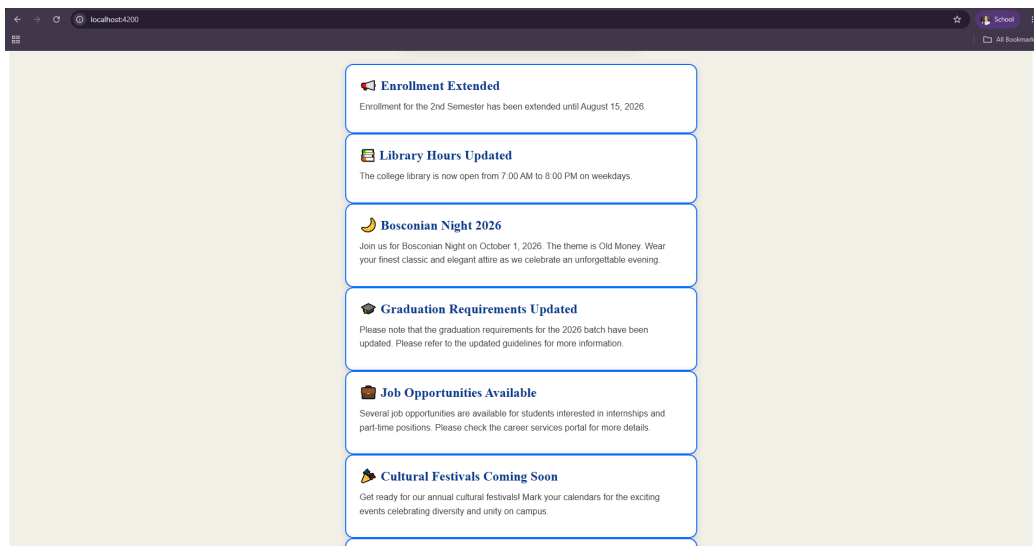
Add **two more announcements** to the array.

You should **not change your HTML**.

If two additional cards automatically appear:

🚩 **FLAG CAPTURED**

📸 **Capture:** Browser showing all ten announcements.



FLAG 6 — Make the Cards More Realistic

Your announcement objects currently contain only:

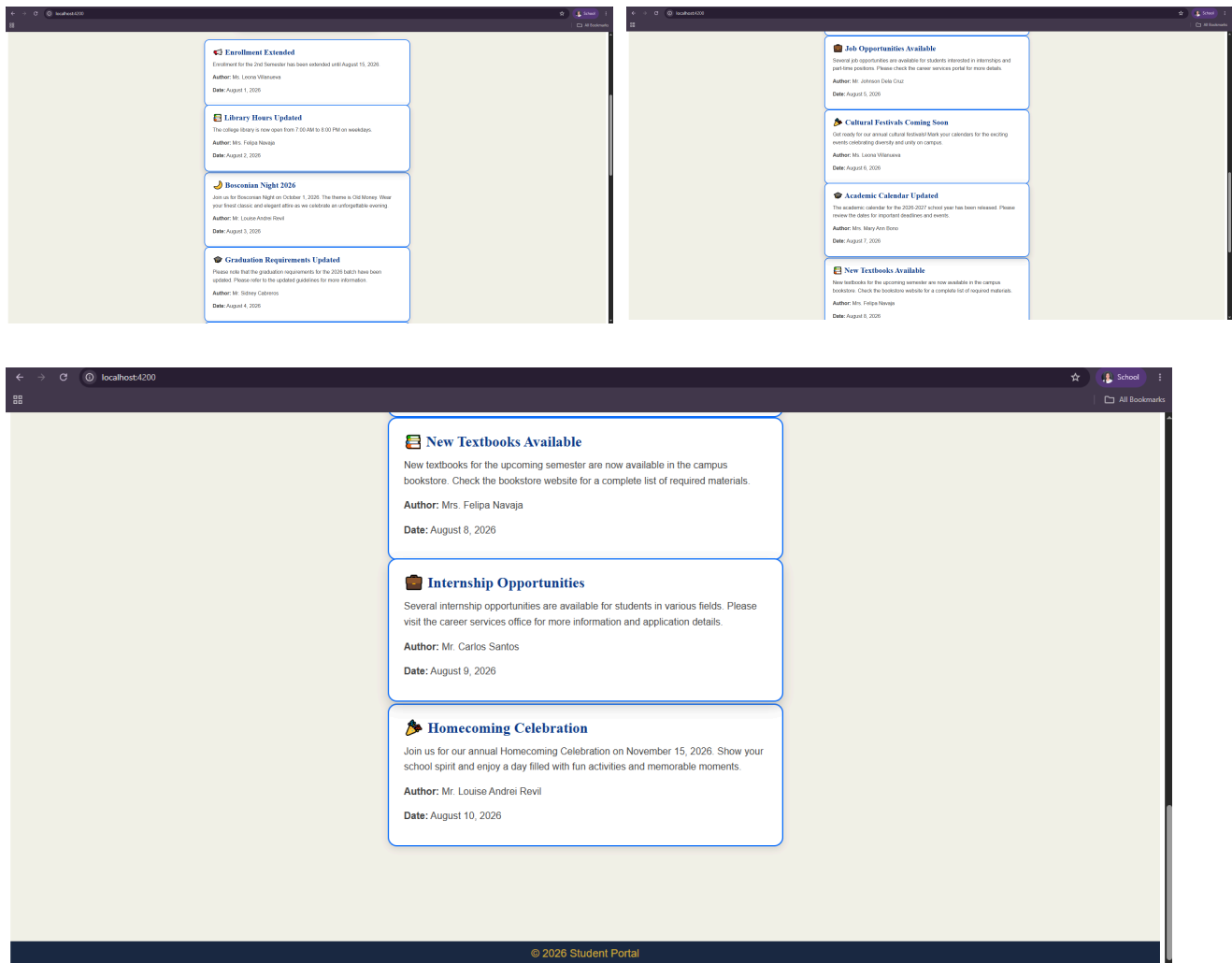
title
body

Add more information: author and date.

Update your card to display all fields.

Use **interpolation** for all values.

 **Capture:** Your richer announcement cards.



FLAG 7 — Student List

Announcements aren't the only thing that can be repeated.

Create another array named students.

Create a simple student card.

Use ***ngFor** to display every student.

Expected example:

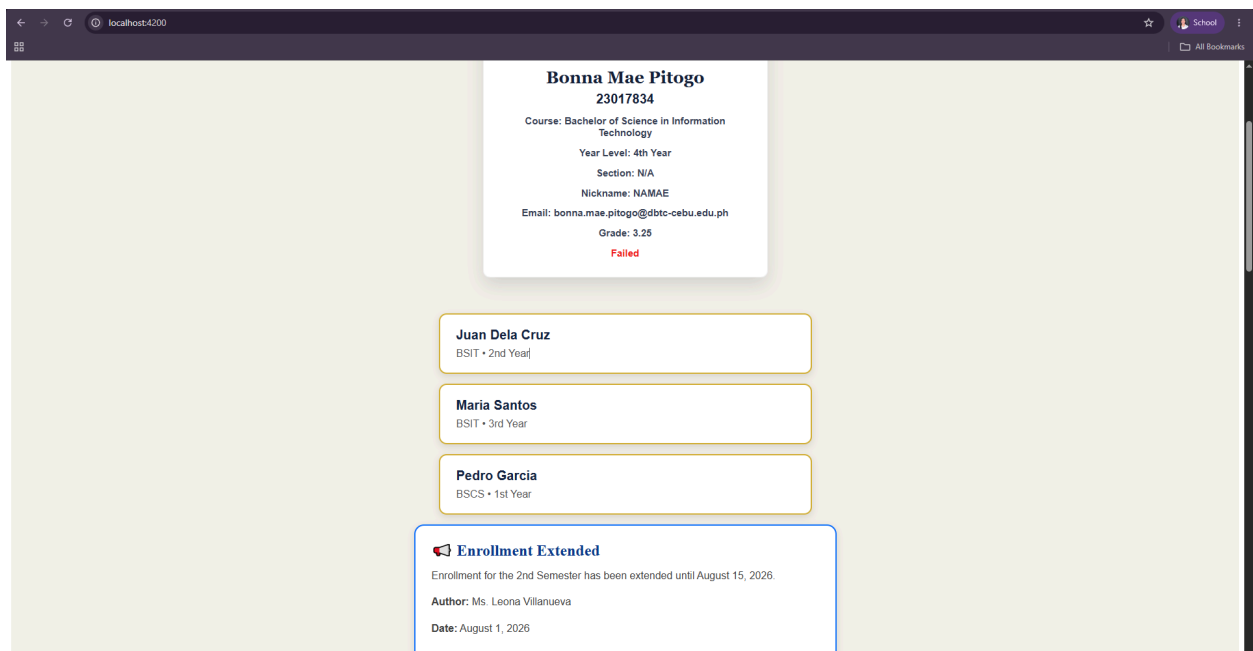
Juan Dela Cruz
BSIT • 2nd Year

Maria Santos
BSIT • 3rd Year

Pedro Garcia
BSCS • 1st Year



Capture: Student list.



FLAG 8 — One Template, Many Students

Add another student to your array.

Do **not** modify the HTML.

The new student should automatically appear.

This is the important concept:

DATA CHANGES



ARRAY CHANGES



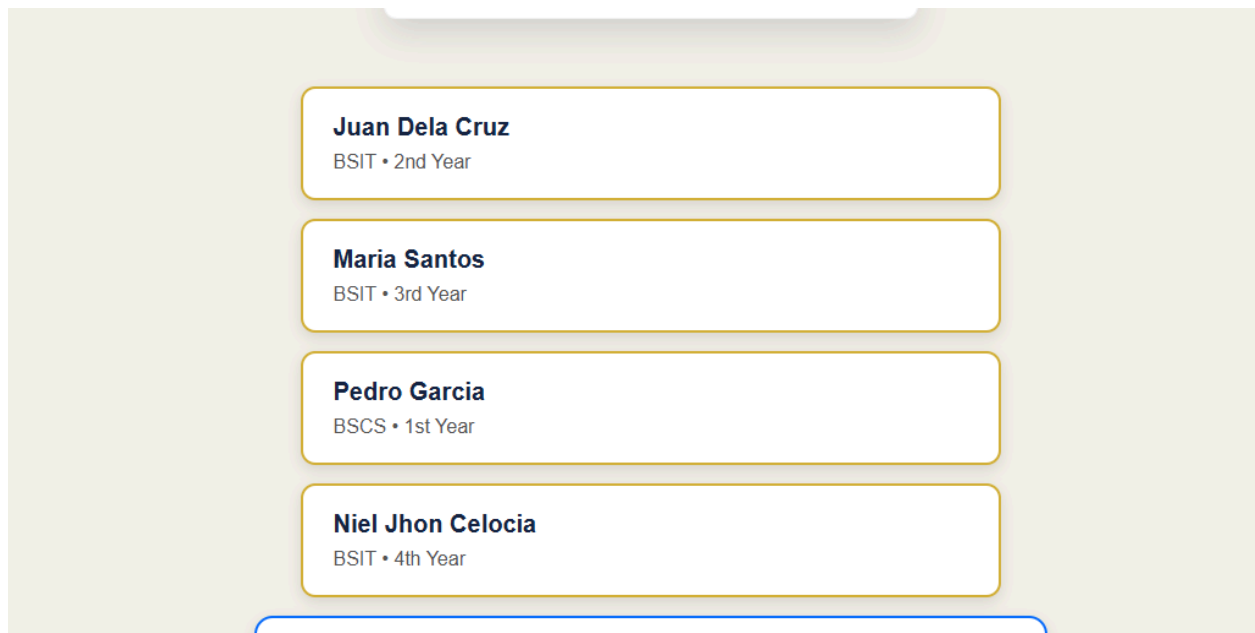
ANGULAR GENERATES THE UI



Capture: Before and after adding the student.

```
TS student-card.ts U • <> student-card.html U # student-card.css U TS app.ts M <> app.html M
src > app > student-card > TS student-card.ts > StudentCard > students
  9   styleUrls: ['./student-card.css']
 10 })
 11 export class StudentCard {
 12
 13     students = [
 14         {
 15             name: 'Juan Dela Cruz',
 16             course: 'BSIT',
 17             year: '2nd Year'
 18         },
 19         {
 20             name: 'Maria Santos',
 21             course: 'BSIT',
 22             year: '3rd Year'
 23         },
 24         {
 25             name: 'Pedro Garcia',
 26             course: 'BSCS',
 27             year: '1st Year'
 28         }
 29     ];
 30
 31 }
```

```
TS student-card.ts U X <> student-card.html U # student-card.css U TS app.ts M <> app.html M
src > app > student-card > TS student-card.ts > StudentCard > students
9   styleUrls: ['./student-card.css']
10 })
11 export class StudentCard {
12
13   students = [
14     {
15       name: 'Juan Dela Cruz',
16       course: 'BSIT',
17       year: '2nd Year'
18     },
19     {
20       name: 'Maria Santos',
21       course: 'BSIT',
22       year: '3rd Year'
23     },
24     {
25       name: 'Pedro Garcia',
26       course: 'BSCS',
27       year: '1st Year'
28     },
29     {
30       name: 'Niel Jhon Celocia',
31       course: 'BSIT',
32       year: '4th Year'
33     }
34   ];
35
36 }
```



FLAG 9 — Research Challenge

I will **not** provide the next syntax.

Research:

**Angular ngFor index*

Find out how to access the position of the current item.

Use it to display:

#1 Juan Dela Cruz

#2 Maria Santos

#3 Pedro Garcia

Without hardcoding the numbers.

Hint:

let i = index



Capture:

- Your numbered student list.
- Your code

```
TS student-card.ts U  student-card.html U X  # student-card.css U  TS app.ts M  app.html M
src > app > student-card > student-card.html > div.student-card
1  <div *ngFor="let student of students; let i = index" class="student-card">
2    <h3>#{{ i + 1 }} {{ student.name }}</h3>
3    <p>{{ student.course }} • {{ student.year }}</p>
4  </div>
```

#1 Juan Dela Cruz

BSIT • 2nd Year

#2 Maria Santos

BSIT • 3rd Year

#3 Pedro Garcia

BSCS • 1st Year

#4 Niel Jhon Celocia

BSIT • 4th Year



Enrollment Extended

Enrollment for the 2nd Semester has been extended until August 15, 2026.

FLAG 10 — Empty List

Now test what happens when there are no announcements.

Change:

```
announcements = [];
```

What happens?

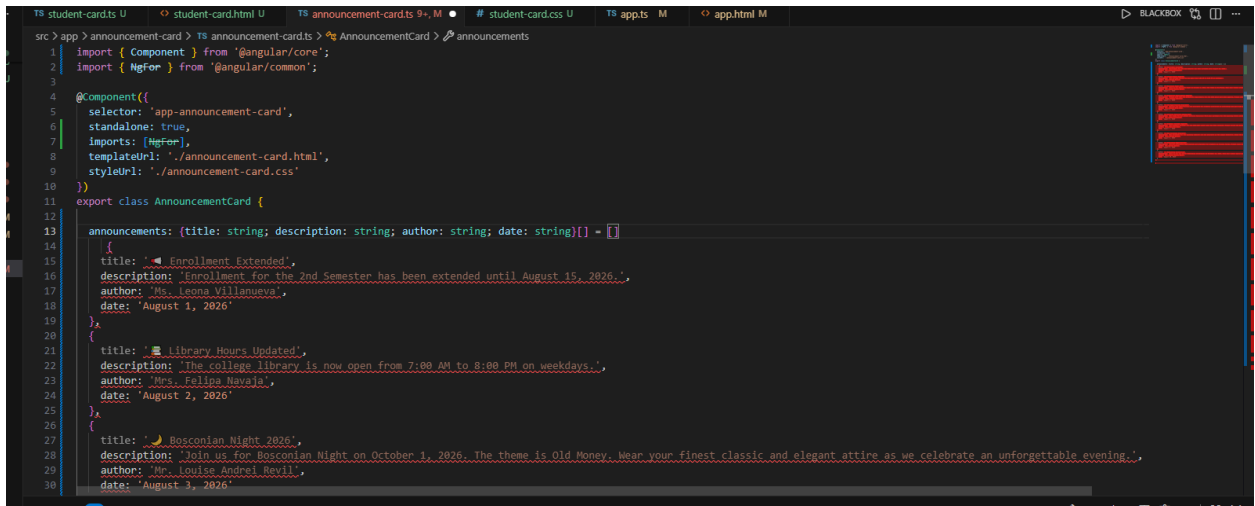
Write a short observation in your documentation:

What happens when the array is empty?

Answer: Error, no items to loop through and no announcement will be displayed.

Then restore your announcements.

 **Capture:** Your experiment and observation.



```
src > app > announcement-card > TS announcement-card.ts > AnnouncementCard > announcements
1 import { Component } from '@angular/core';
2 import { NgFor } from '@angular/common';
3
4 @Component({
5   selector: 'app-announcement-card',
6   standalone: true,
7   imports: [NgFor],
8   templateUrl: './announcement-card.html',
9   styleUrls: ['./announcement-card.css']
10 })
11 export class AnnouncementCard {
12
13   announcements: {title: string; description: string; author: string; date: string}[] = [
14     {
15       title: '📢 Enrollment Extended',
16       description: 'Enrollment for the 2nd Semester has been extended until August 15, 2026.',
17       author: 'Ms. Leona Villanueva',
18       date: 'August 1, 2026'
19     },
20     {
21       title: '📖 Library Hours Updated',
22       description: 'The college library is now open from 7:00 AM to 8:00 PM on weekdays.',
23       author: 'Mrs. Felina Navaid',
24       date: 'August 2, 2026'
25     },
26     {
27       title: '🌃 Bosconian Night 2026',
28       description: 'Join us for Bosconian Night on October 1, 2026. The theme is Old Money. Wear your finest classic and elegant attire as we celebrate an unforgettable evening.',
29       author: 'Mr. Louise Andrei Revil',
30       date: 'August 3, 2026'
31     }
32   ]
33 }
```

FLAG 11 — Dynamic Student Portal

Your Student Portal should now contain:

Your Student Profile

Still using interpolation:

Name

Student ID

Course

Year

Grade

Academic Standing

Email

Announcements

Generated from:

announcements = [...]

Using:

*ngFor

Student List

Generated from:

students = [...]

Using:

*ngFor and index

Your final requirement

Final Screenshot

- Entire browser UI of all requirements in this activity

localhost:4200

School

All Bookmarks

DON BOSCO STUDENT PORTAL



Bonna Mae Pitogo
23017834
Course: Bachelor of Science in Information Technology
Year: 4th Year
Grade: 3.25
Failed
Academic Standing: Good Standing
Email: bonna.mae.pitogo@dbtc-cebu.edu.ph

ANNOUNCEMENTS:

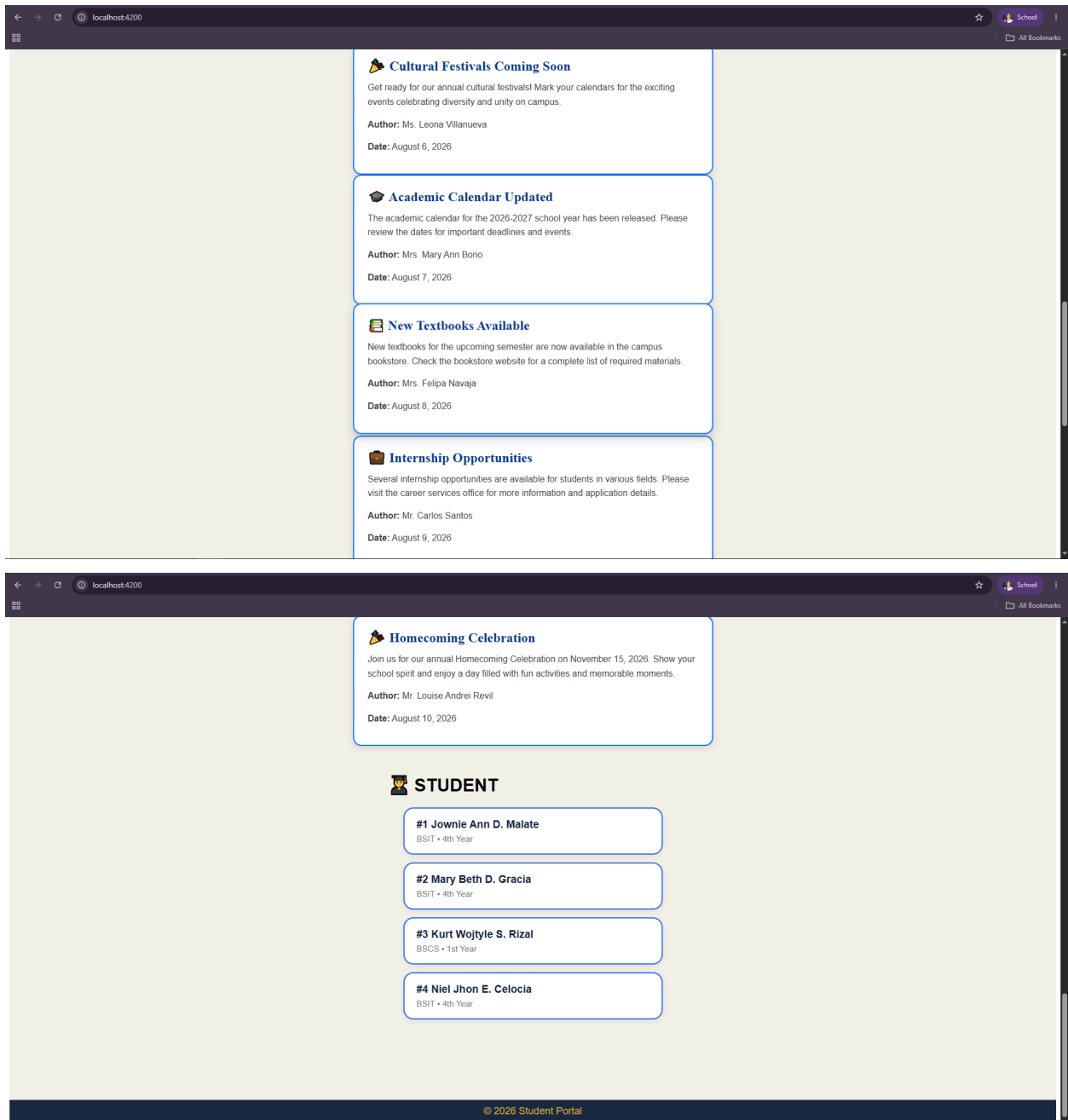
 **Enrollment Extended**
Enrollment for the 2nd Semester has been extended until August 15, 2026.
Author: Ms. Leona Villanueva
Date: August 1, 2026

 **Library Hours Updated**
The college library is now open from 7:00 AM to 8:00 PM on weekdays.
Author: Mrs. Felipa Navaja
Date: August 2, 2026

 **Bosconian Night 2026**
Join us for Bosconian Night on October 1, 2026. The theme is Old Money. Wear your finest classic and elegant attire as we celebrate an unforgettable evening.
Author: Mr. Louise Andrei Revil
Date: August 3, 2026

 **Graduation Requirements Updated**
Please note that the graduation requirements for the 2026 batch have been updated. Please refer to the updated guidelines for more information.
Author: Engr. Sidney Cabrerros
Date: August 4, 2026

 **Job Opportunities Available**
Several job opportunities are available for students interested in internships and part-time positions. Please check the career services portal for more details.
Author: Mr. Johnson Dela Cruz



FLAG 12 — Commit & Push

📸 Final Screenshot

- Successful push to feature/ngFor

